

SIMULATION EXPERIMENT-02

NUMERICAL SIMULATION AND PERFORMANCE ANALYSIS OF MICROWAVE TRANSMISSION LINES AND TWO-PORT NETWORKS USING SCATTERING PARAMETERS

→ Objectives of this Experiment:

1. To model coaxial, waveguide or microstrip transmission structures for microwave-frequency operation.
2. To determine characteristic impedance, propagation constant and transmission characteristics of the selected structure.
3. To obtain and analyze S11 , S21 , S12 and S22 over a specified frequency range.
4. To evaluate return loss, insertion loss and transmission efficiency from the simulated S-parameter response.

PROGRAM

```
clc;

clear;

close all;

f = linspace(1e9,10e9,501);

c = 3e8;

Z0 = 50;

ZL = 75;

L = 0.05;

alpha = 0.05;

vf = 0.66;

lambda = c./(f*vf);

beta = 2*pi./lambda;

gamma = alpha + 1j*beta;

Zin = Z0.*(ZL + Z0.*tanh(gamma*L))./(Z0 + ZL.*tanh(gamma*L));

S11 = (Zin-Z0)./(Zin+Z0);

S22 = S11;

S21 = exp(-gamma*L);

S12 = S21;
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RL = -20*log10(abs(S11));
IL = -20*log10(abs(S21));
VSWR = (1+abs(S11))./(1-abs(S11));
fprintf('-----\n');
fprintf(' MICROWAVE TWO-PORT NETWORK ANALYSIS\n');
fprintf('-----\n');
fprintf('Reference Impedance = %.2f Ohm\n',Z0);
fprintf('Load Impedance    = %.2f Ohm\n',ZL);
fprintf('Transmission Line  = %.2f cm\n',L*100);
fprintf('Velocity Factor    = %.2f\n',vf);
fprintf('S-PARAMETERS AT 5 GHz\n');
fprintf('-----\n');
[~,idx] = min(abs(f-5e9));
fprintf('S11 = %.4f %+.4fj\n',real(S11(idx)),imag(S11(idx)));
fprintf('S21 = %.4f %+.4fj\n',real(S21(idx)),imag(S21(idx)));
fprintf('S12 = %.4f %+.4fj\n',real(S12(idx)),imag(S12(idx)));
fprintf('S22 = %.4f %+.4fj\n',real(S22(idx)),imag(S22(idx)));
fprintf('PERFORMANCE AT 5 GHz\n');
fprintf('-----\n');
fprintf('Return Loss  = %.3f dB\n',RL(idx));
fprintf('Insertion Loss = %.3f dB\n',IL(idx));
fprintf('VSWR        = %.3f\n',VSWR(idx));
fprintf('-----\n');
fprintf('SIMULATION COMPLETED SUCCESSFULLY\n');
fprintf('-----\n');
figure;
plot(f/1e9,20*log10(abs(S11)),'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');

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ylabel('|S11| (dB)');
title('Reflection Coefficient S11');
figure;
plot(f/1e9,20*log10(abs(S21)),'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('|S21| (dB)');
title('Transmission Coefficient S21');
figure;
plot(f/1e9,RL,'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Return Loss (dB)');
title('Return Loss');
figure;
plot(f/1e9,IL,'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('Insertion Loss (dB)');
title('Insertion Loss');
figure;
plot(f/1e9,VSWR,'LineWidth',2);
grid on;
xlabel('Frequency (GHz)');
ylabel('VSWR');
title('VSWR vs Frequency');
figure;
plot(f/1e9,abs(S11),'LineWidth',2);
hold on;

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plot(f/1e9,abs(S21),'LineWidth',2);

plot(f/1e9,abs(S12),'LineWidth',2);

plot(f/1e9,abs(S22),'LineWidth',2);

grid on;

xlabel('Frequency (GHz)');

ylabel('Magnitude');

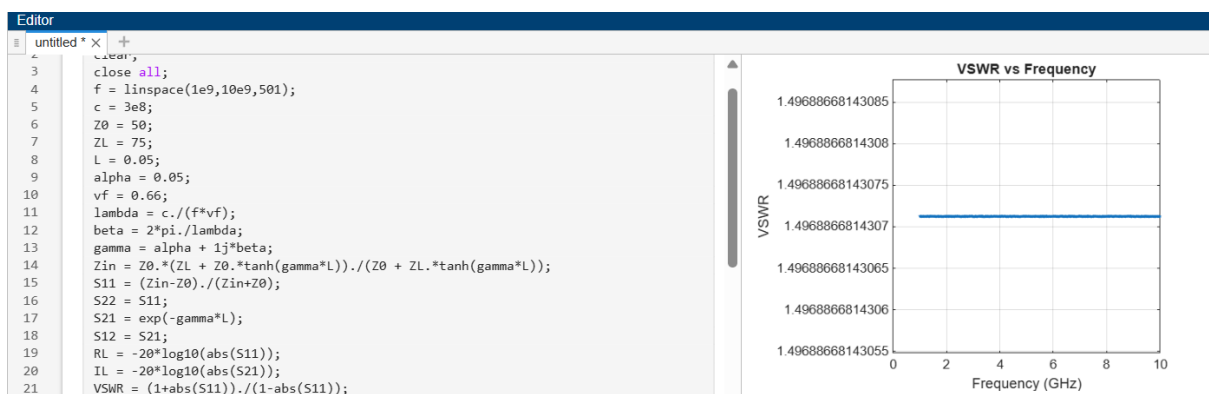
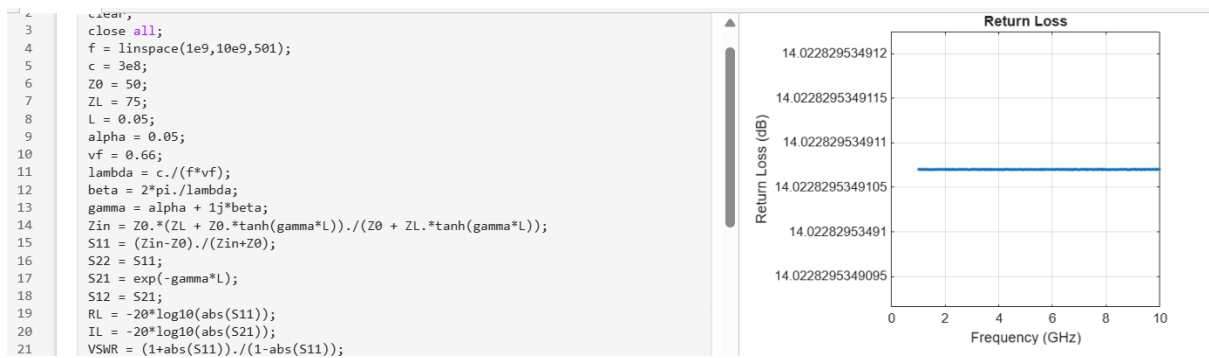
title('S-Parameter Magnitudes');

legend('S11','S21','S12','S22');

```

OUTPUT





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Zin = Z0.*(ZL + Z0.*tanh(gamma*L))./(Z0 + ZL.*tanh(gamma*L));
S11 = (Zin-Z0)./(Zin+Z0);
S22 = S11;
S21 = exp(-gamma*L);
S12 = S21;
RL = -20*log10(abs(S11));
IL = -20*log10(abs(S21));
VSWR = (1+abs(S11))./(1-abs(S11));
fprintf('-----\n');
fprintf(' MICROWAVE TWO-PORT NETWORK ANALYSIS\n');
fprintf('-----\n');
fprintf('Reference Impedance = %.2f Ohm\n',Z0);
fprintf('Load Impedance      = %.2f Ohm\n',ZL);
fprintf('Transmission Line   = %.2f cm\n',L*100);
fprintf('Velocity Factor      = %.2f\n',vf);
fprintf('S-PARAMETERS AT 5 GHz\n');
fprintf('-----\n');
[~,idx] = min(abs(f-5e9));
fprintf('S11 = %.4f %+.4fj\n',real(S11(idx)),imag(S11(idx)));

```

